

Learning Objective 3.4: Calculate the phase weights required to steer a beam to a given angle and predict the resulting array pattern.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **From geometry to phase.** The course array is a uniform linear array of $N = 8$ patch elements spaced $d = 14$ mm apart, receiving at $f = 10.3$ GHz. A plane wave arrives $\theta_0 = 25^\circ$ from broadside.

- (a) In one or two sentences, explain which element the wavefront reaches first and why a fixed phase applied to each element can make all eight received signals add together.

- (b) Compute the wavelength, the electrical spacing d/λ , and the product kd in degrees.

$kd =$

- (c) Compute the progressive element-to-element phase $\Delta\phi$ required to steer the beam to 25° .

$\Delta\phi =$

- (d) Compute the arrival-time difference between neighboring elements at this steer angle, in picoseconds.

$\Delta t =$

2. **The phase table.** The same array is commanded to $\theta_0 = -40^\circ$. Element 0 is the phase reference, and the ADAR1000 accepts a phase setting between 0° and 360° .

(a) Compute $\Delta\phi$ for this steer angle, including its sign.

$$\Delta\phi =$$

(b) Compute the commanded phase of element 5 from $\phi_n = -n\Delta\phi$, before any wrapping.

$$\phi_5 =$$

(c) Give the value actually written to element 5, wrapped into 0° to 360° .

$$\phi_5 =$$

(d) Wrapping changed the number by 360° and did not change the pattern. Explain why in one or two sentences, and state what information the wrapped table no longer carries.

3. **The steered pattern.** The array is steered to $\theta_0 = 50^\circ$. Use $\psi = kd(\sin(\theta) - \sin(\theta_0))$ with $kd = 173.2^\circ$, and recall $\lambda/Nd = 29.1/112 = 0.260$.

- (a) Compute the array-factor argument ψ at the observation angle $\theta = 30^\circ$.

$\psi =$

- (b) Locate the first null on the broadside side of the main lobe from $\sin(\theta) = \sin(\theta_0) - \lambda/Nd$.

$\theta =$

- (c) The matching null on the other side requires $\sin(\theta) = 0.766 + 0.260 = 1.026$. Explain what that result means physically and what the pattern looks like on that side of the beam.
- (d) Two commanded angles differ by 5° near broadside, and two others differ by 5° near 60° . Which pair requires the larger change in $\Delta\phi$, and what does that mean for an array searching a sector in equal phase steps?

4. **Scanning costs beamwidth.** Continue with $N = 8$, $d = 14$ mm, $\lambda = 29.1$ mm. Use $\theta_{\text{HP}} \approx 0.886\lambda/(Nd\cos(\theta_0))$ and the broadside uniform-array directivity $D \approx 2Nd/\lambda$.

- (a) Compute the broadside half-power beamwidth of the array.

$$\theta_{\text{HP}} =$$

- (b) Compute the half-power beamwidth when the beam is steered to $\theta_0 = 55^\circ$.

$$\theta_{\text{HP}} =$$

- (c) Compute the broadside directivity of the array in dBi . Then, using the fact that the effective aperture shrinks to its projection $Nd\cos(\theta_0)$ when the beam is steered, compute the expected gain reduction at $\theta_0 = 55^\circ$ and name the effect.

$$D(0) =$$

$$\Delta G =$$

- (d) A tracking requirement allows the beam to broaden to at most 20° . Find the largest usable scan angle, and state the trade-off this puts on the system design in one sentence.

$$\theta_0 =$$

5. **The inverse problem.** You read the following phase settings out of the beamformer. The array is the course array, $kd = 173.2^\circ$.

n	0	1	2	3	4	5	6	7
ϕ_n (deg)	0	260.7	161.4	62.1	322.8	223.5	124.2	24.9

- (a) Difference neighboring elements, unwrap, and report the common step and the value of $\Delta\phi$.

$\Delta\phi =$

- (b) Recover the commanded steer angle.

$\theta_0 =$

- (c) Suppose the seven differences had come out as -99.3° for six of them and -84° for one. State two things that could produce that result, and what you would check first.

- (d) The ADAR1000 sets phase on a 2.8125° grid. Give the value element 1 actually takes and the resulting error.

$\phi_1 =$

Documentation: _____